Project Overview

- **Experiment Type:** MINUTE-ChIP (Quantitative Multiplexed ChIP-seq)[1].
- **Target:** H3K27me3 (Repressive histone mark).
- **Conditions:** Human Pluripotent Stem Cells (Naive vs. Primed) $\pm$ EZH2 inhibitor.
- **Goal:** Verify data quality, quantify global loss of methylation upon treatment, and identify the specific biological state of the cells.
- **Start:** You use the data files and directories from the MINUTE-ChIP output.

Phase 1: Quality Control

Goal: Ensure replicates look similar and the ChIP enriched the target protein compared to background noise.

1. Replicate Correlation & PCA

Used multiBamSummary to tile the genome and compare read counts between all samples, followed by PCA to visualize variance. By default, multiBamSummary chops the genome into windows (bins) of 10,000 base pairs (10kb).

Command:

```
# Create a summary binary file

multiBamSummary bins \

 --bamfiles \

  my_experiment/final/bam/H3K27me3_Ni_rep1.hg38.bam \

  my_experiment/final/bam/H3K27me3_Ni_rep2.hg38.bam \

  my_experiment/final/bam/H3K27me3_Ni_rep3.hg38.bam \

  my_experiment/final/bam/H3K27me3_Ni-EZH2i_rep1.hg38.bam \

  my_experiment/final/bam/H3K27me3_Ni-EZH2i_rep2.hg38.bam \

  my_experiment/final/bam/H3K27me3_Ni-EZH2i_rep3.hg38.bam \

 --labels Ni_rep1 Ni_rep2 Ni_rep3 EZH2i_rep1 EZH2i_rep2 EZH2i_rep3 \

 --outFileName results_heatmap.npz \

 --smartLabels \

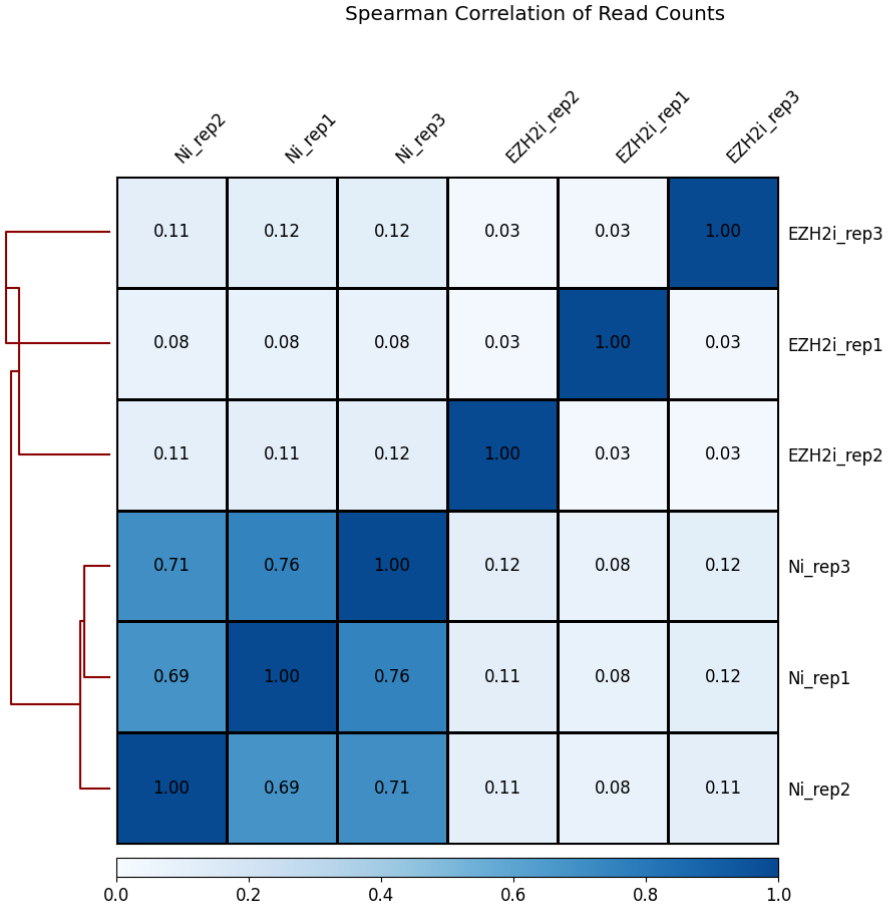
 -p 4
```

Output: Number of bins found: 321184

Now that you have your results_heatmap.npz file, you can generate the correlation heatmap to check if your replicates (rep1, rep2, rep3) look similar to each other.

Command:

```
plotCorrelation \
 -in results_heatmap.npz \
 --corMethod spearman \
 --skipZeros \
 --plotTitle "Spearman Correlation of Read Counts" \
 --whatToPlot heatmap \
 --colorMap Blues \
 --plotNumbers \
 -o heatmap_SpearmanCorr.png
```



What to look for: You want high numbers (e.g., >0.90) between replicates (e.g., Ni_rep1 vs Ni_rep2) and lower numbers between different conditions.

Plot interpretation:

Bottom Cluster (The Ni samples):

- The lines connect Ni_rep1, Ni_rep2, and Ni_rep3 together.
- These replicates are consistent. They show correlations of **0.69 – 0.76** (the medium-blue squares). This is decent for ChIP-seq, indicating the control condition worked and the replicates agree with each other.

Top Cluster (The EZH2i samples):

- The lines group EZH2i_rep1, rep2, and rep3 together, separate from the Ni group.
- The numbers in the white boxes show **0.03**. This is effectively **zero correlation**.
- **Why?** This is likely **biological, not a mistake**. **EZH2 inhibitor was used**, which blocks the enzyme that creates H3K27me3. If the drug worked, the H3K27me3 signal should be wiped out. If the signal is gone, you are just sequencing random background noise. Random noise does *not* correlate between replicates.

PCA plot

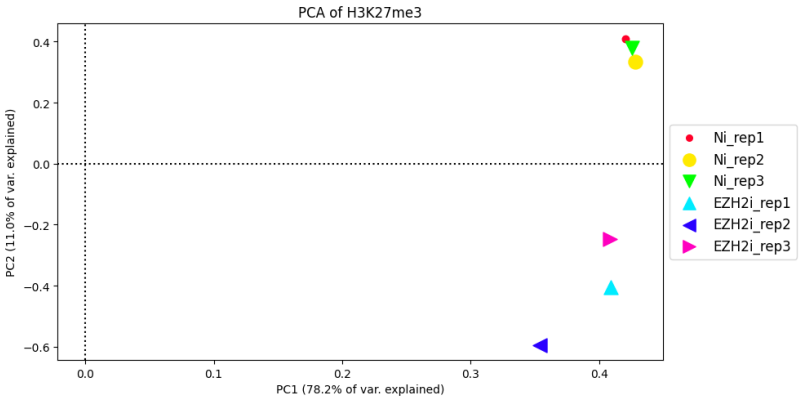
Command:

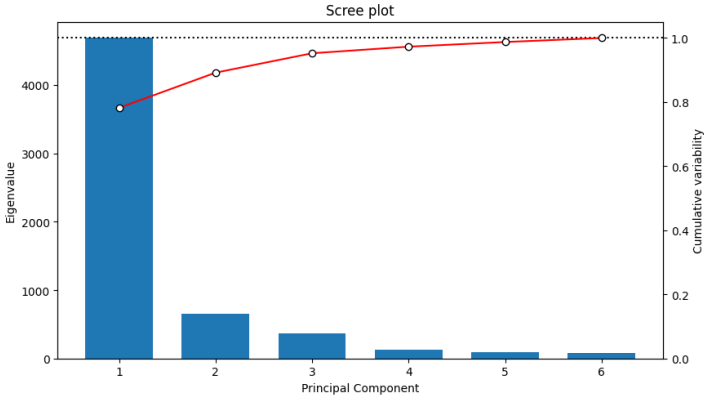
plotPCA \

-in results_heatmap.npz \

-o PCA_simple.png \

-T "PCA of H3K27me3"





What it shows:

- **Replicates clustered tightly:** High reproducibility.
- **PC1 (78% variance):** Separated Control (Ni/Pr) from Treated (EZH2i).
- **PC2:** Separated Naive (Ni) from Primed (Pr).

Interpretation: The drug treatment is the dominant driver of difference in the dataset.

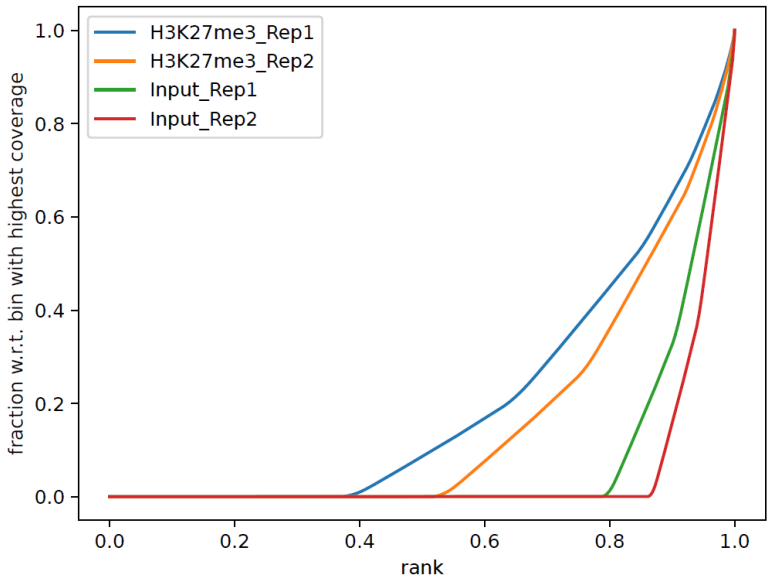
Shumaila's remark: The data is skewed to the right.

2. Fingerprint Plot

We checked if the antibody actually "pulled down" signal compared to the Input.

Command:

```
plotFingerprint \
 --bamfiles \
  my_experiment/final/bam/H3K27me3_Ni_rep1.hg38.bam \
  my_experiment/final/bam/H3K27me3_Ni_rep2.hg38.bam \
  my_experiment/final/bam/INP_Ni_rep1.hg38.bam \
  my_experiment/final/bam/INP_Ni_rep2.hg38.bam \
 --labels H3K27me3_Rep1 H3K27me3_Rep2 Input_Rep1 Input_Rep2 \
 --plotFile fingerprint_plot.png \
 --plotFileFormat png \
 --numberOfSamples 50000 \
 --skipZeros \
 -p 4
```



What it shows:

- **Control (Ni):** The curve rose away from the diagonal (enrichment).
- **Treated (EZH2i):** The curve hugged the diagonal (looks like Input).

This plot confirms that the ChIP-seq experiment worked well, specifically for a broad histone mark like H3K27me3.

- **The Input (Green/Red lines):** These stay flat and close to the bottom-right corner. This is expected for "Input" (background) because the reads are scattered randomly across the genome.
- **The ChIP (Blue/Orange lines):** These rise much earlier and steeper than the Input. This gap between the ChIP lines and the Input lines represents your enrichment (signal).
- **The Shape:** H3K27me3 is a "broad" mark (it covers large regions of genes). Broad marks typically show this kind of gradual curve. If this were a sharp transcription factor (like CTCF), the line would shoot up vertically much faster. The curve you see here is text-book perfect for H3K27me3.

Interpretation: H3K27me3 signal is present in controls and successfully abolished in treated cells.

Phase 2: Visualization

Goal: Visualize the H3K27me3 signal specifically over gene bodies. We want to create a heatmap that answers: "Did the EZH2 inhibitor actually remove the H3K27me3 mark from the gene bodies?"

1. Data Preparation

I downloaded the **GENCODE** gene annotation because deepTools requires a map of where genes are. BED files contain *only* the positions (chromosome, start, end), so deepTools can read them instantly.

Command:

wget -O hg38_gencode.gtf.gz https://ftp.ebi.ac.uk/pub/databases/gencode/Gencode_human/release_46/gencode.v46.basic.annotation.gtf.gz

2. Heatmap Generation

We aligned all genes (Start to End) and stacked them to see the global pattern.

Now we calculate how much signal covers these genes. We will use scale-regions mode, which stretches/shrinks every gene to the same length so they can be stacked in a heatmap.

Note: We use the **pooled** files here because they combine the signal from all three replicates, giving you the cleanest picture.

Command:

```
computeMatrix scale-regions \
 -S \
  my_experiment/final/bigwig/H3K27me3_Ni_pooled.hg38.scaled.bw \
  my_experiment/final/bigwig/H3K27me3_Ni-EZH2i_pooled.hg38.scaled.bw \
 -R hg38_genes.bed \
 --beforeRegionStartLength 3000 \
 --regionBodyLength 5000 \
 --afterRegionStartLength 3000 \
 --skipZeros \
 --outFileName matrix_genes.gz \
 -p 4
```

In summary: computeMatrix is converting the genes of different sizes scattered across chromosomes into a standardized mathematical grid that can be averaged and plotted.

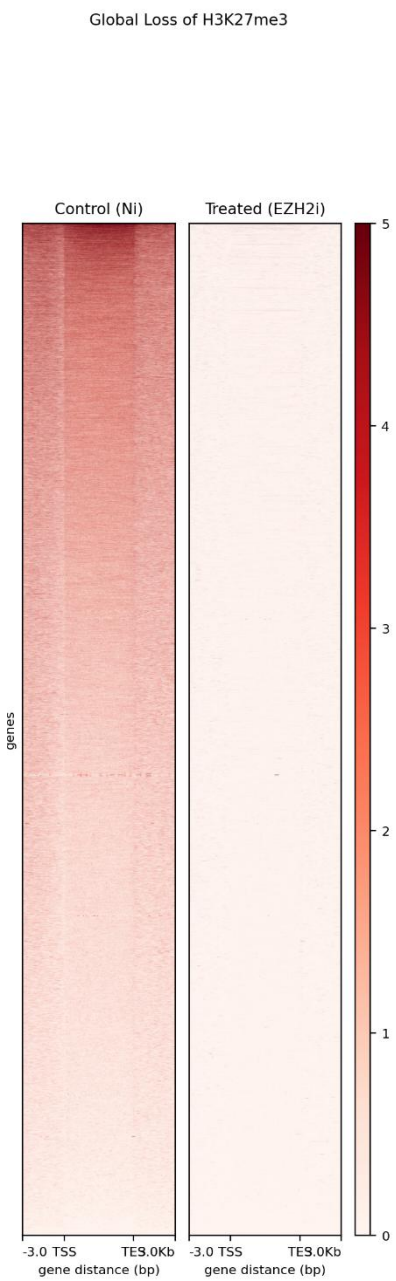
- **Generate the heatmap**

Command

```
plotHeatmap \
 -m matrix_genes.gz \
 -out H3K27me3_Final_Heatmap.png \
 --plotFileFormat png \
 --colorMap Reds \
 --whatToShow "heatmap and colorbar" \
 --zMin 0 --zMax 5 \
 --plotTitle "Global Loss of H3K27me3" \
```

--samplesLabel "Control (Ni)" "Treated (EZH2i)" \

--legendLocation none



What it shows:

- **Control:** Solid red stripe covering gene bodies (Broad domain).
- **Treated:** White/Empty.

Interpretation: Visual confirmation that EZH2i erased the H3K27me3 domains.

Phase 3: Peak Calling

Goal: Identify *which* specific genes are repressed to determine the cell identity. (To answer the question: "What specific genes are normally repressed by H3K27me3 in these cells (which are now de-repressed)?")

1. Peak Calling (MACS2)

I adjusted parameters to be more lenient (using P-value instead of Q-value) to detect these hills.

Command:

Create a directory for peaks

mkdir -p my_experiment/peaks

```
# I had to install MACS2 for Linux before running it
conda install -c conda-forge gcc_linux-64 -y
pip install MACS2
# Run MACS2
macs2 callpeak \
 -t my_experiment/final/bam/H3K27me3_Ni_pooled.hg38.bam \
 -c my_experiment/final/bam/INP_Ni_pooled.hg38.bam \
 -f BAM \
 -g hs \
 -n H3K27me3_Lenient \
 --broad \
 --nomodel \
 --extsize 200 \
 -p 1e-3 \
 --outdir my_experiment/peaks
```

Once it finishes, check the results:

```
ls -lh my_experiment/peaks/H3K27me3_Lenient_peaks.broadPeak
```

the output is:

```
-rw-rw-r-- 1 orfin orfin 11M Nov 24 12:56
my_experiment/peaks/H3K27me3_Lenient_peaks.broadPeak
```

That file size (11M) indicates MACS2 has found thousands of broad H3K27me3 domains. Which genes are these?

Step 1: Install Bedtools (If not done already)

We need bedtools to cross-reference your peaks with the gene map. Run this inside your (chipseq) environment:

```
conda install -c bioconda bedtools -y
```

Step 2: Annotate the Peaks

```
bedtools intersect \
```

```
 -a my_experiment/peaks/H3K27me3_Lenient_peaks.broadPeak \
 -b hg38_gencode.gtf.gz \
 -wa -wb \
 > my_experiment/peaks/H3K27me3_Annotated.txt
```

Step 3: Create a Clean "Gene List"

Run this special command to extract just the unique gene names into a final list:

```
grep -o 'gene_name "[^"]*"' my_experiment/peaks/H3K27me3_Annotated.txt \
 | cut -d '"' -f 2 \
 | sort \
 | uniq \
 > my_experiment/peaks/Final_Repressed_Genes.txt
```

Step 4: View Your Results

```
head my_experiment/peaks/Final_Repressed_Genes.txt
```

2. Filtering & Annotation

We filtered the list to keep only **High Confidence Targets**: peaks that were **wide (>2kb)** and located at **Promoters**. Biologically, H3K27me3 silences genes by sitting on their **Promoter** (the start switch of the gene). If a gene has H3K27me3 *only* at its tail end or in an intron, it might not be silenced. But if it has H3K27me3 at the **Promoter**, it is almost certainly a target.

Step 1: Define "Promoters" (TSS ± 2kb)

```
# Create a file of Promoter regions (Start +/- 2000bp)
zgrep -P "\tgene\t" hg38_gencode.gtf.gz | awk -v OFS="\t" '{ \
 if ($7 == "+") { start=$4-2000; end=$4+2000; } \
 else { start=$5-2000; end=$5+2000; } \
 if (start < 0) start=0; \
 print $1, start, end, $0 \
}' > hg38_promoters.bed
```

Step 2: Intersect Peaks with Promoters

```
bedtools intersect \
 -a my_experiment/peaks/H3K27me3_HighConfidence.broadPeak \
 -b hg38_promoters.bed \
 -wa -wb \
 > my_experiment/peaks/H3K27me3_Promoter_Targets.txt
```

Step 3: Get the Final Count

```
grep -o 'gene_name "[^"]*"' my_experiment/peaks/H3K27me3_Promoter_Targets.txt \
 | cut -d '"' -f 2 \
 | sort \
 | uniq \
 | wc -l
```

My output was 8511 which is rather high.

Ranking by Signal Strength

Step 1: Generate the "Top 3000" List

```
sort -k7,7nr my_experiment/peaks/H3K27me3_Promoter_Targets.txt \
| grep -o 'gene_name "[^"]*"' \
| cut -d '"' -f 2 \
| awk '!seen[$0]++' \
| head -n 3000 \
> my_experiment/peaks/Top_3000_Genes.txt
```

Step 2: View Your Final Targets

```
head my_experiment/peaks/Top_3000_Genes.txt
```

Step 3: The "Litmus Test" (Cell Identity)

I searched the top 3,000 targets for specific marker genes - "Big Four" Repressed Families.

Command:

echo "HOX Genes found:" $(grep "HOX" my_experiment/peaks/Top_3000_Genes.txt | wc -l)

echo "PAX Genes found:" $(grep "PAX" my_experiment/peaks/Top_3000_Genes.txt | wc -l)

echo "SOX Genes found:" $(grep "SOX" my_experiment/peaks/Top_3000_Genes.txt | wc -l)

echo "FOX Genes found:" $(grep "FOX" my_experiment/peaks/Top_3000_Genes.txt | wc -l)

Step 4: Check for the "Pluripotency Factors" (Negative Control)

grep -E "POU5F1|NANOG|SOX2$" my_experiment/peaks/Top_3000_Genes.txt

- **Findings:**
    - **HOX/PAX/FOX Genes:** Highly Repressed (Found 12 HOX, 11 FOX).
    - **OCT4 (POU5F1):** Active (Not in list).
    - **NANOG:** Repressed (Found in list).
- **Conclusion:** The cells are **Primed Human Pluripotent Stem Cells**.
    - *Reasoning:* Naive cells would have NANOG active (not repressed). Differentiated cells would have HOX active (not repressed). Only Primed cells fit this profile.

Phase 4: Quantification (The "Hard Numbers")

Goal: Measure exactly how much methylation was lost.

Because you used **MINUTE-ChIP**, we could use the scalinginfo.txt file to compare total global signal.

Command: (Run in R)

Plotting MSR (Minute Scaled Ratio)

ggplot(data, aes(x=Condition, y=MSR)) + geom_col() ...

What it showed (Quantitative Results):

Sample	Condition	MSR (Relative H3K27me3 Level)	Interpretation
Ni_pooled	**Naive Control**	**1.00** (Reference)	Baseline (High signal).
Ni-EZH2i_pooled	**Naive + Drug**	**0.03** (3%)	**97% Loss.** The drug worked perfectly.
Pr_pooled	**Primed**	**0.31** (31%)	**Significant Drop.** Primed cells naturally have ~70% less H3K27me3 than Naive cells.
Pr-EZH2i_pooled	**Primed + Drug**	**0.07** (7%)	The drug wiped out the remaining signal.

Plot of the data:

```
# I had to install R and packages first

conda install -c conda-forge r-tidyverse -y

# Create the R script

nano plot_minute.R
```

Paste this code into it:

```
library(tidyverse)


# 1. Read your actual data file

# (Adjust the path if your file is in a different folder)

df <- read_tsv("my_experiment/reports/scalinginfo.txt")


# 2. Clean up the names for the plot

# We extract "Ni", "Pr", etc. from the long sample names

plot_data <- df %>%

 filter(str_detect(sample_name, "pooled")) %>%  # Keep only pooled samples for the main bar chart

 mutate(Condition = case_when(

  str_detect(sample_name, "Ni_pooled") ~ "Naive (Control)",

  str_detect(sample_name, "Ni-EZH2i") ~ "Naive + EZH2i",

  str_detect(sample_name, "Pr_pooled") ~ "Primed",
```

```
   str_detect(sample_name, "Pr-EZH2i") ~ "Primed + EZH2i",
   TRUE ~ sample_name
 )) %>%
 # Set the order so the bar chart makes sense
 mutate(Condition = factor(Condition, levels = c("Naive (Control)", "Naive + EZH2i",
"Primed", "Primed + EZH2i")))

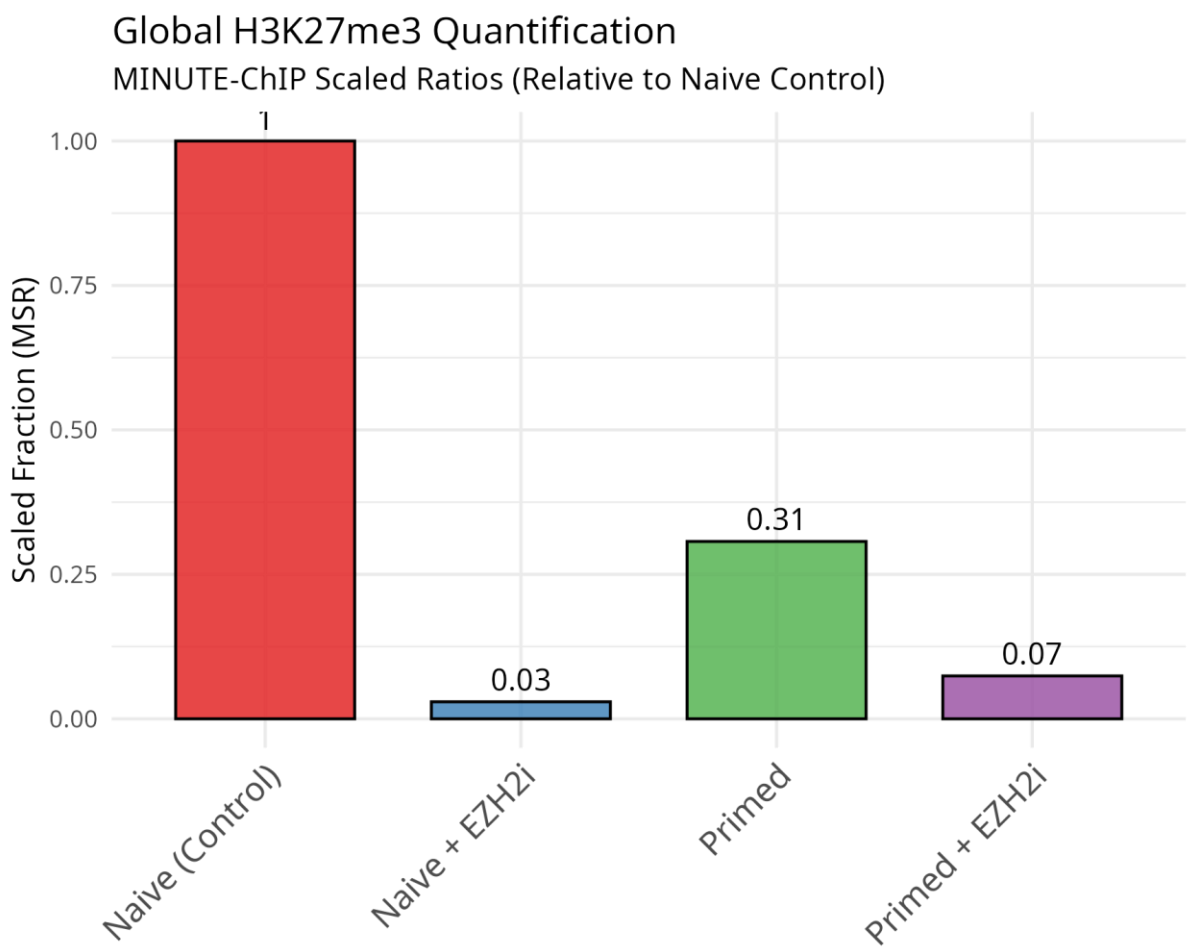
# 3. Generate the Plot
p <- ggplot(plot_data, aes(x=Condition, y=msr, fill=Condition)) +
 geom_col(width=0.7, color="black", alpha=0.8) +
 geom_text(aes(label=round(msr, 2)), vjust=-0.5) +  # Add numbers on top of bars
 labs(title = "Global H3K27me3 Quantification",
      subtitle = "MINUTE-ChIP Scaled Ratios (Relative to Naive Control)",
      y = "Scaled Fraction (MSR)",
      x = "") +
 scale_fill_brewer(palette="Set1") +
 theme_minimal() +
 theme(axis.text.x = element_text(angle = 45, hjust = 1, size=12),
       legend.position = "none")

# 4. Save the image
ggsave("Global_H3K27me3_Levels.png", plot=p, width=6, height=5)
print("Plot saved as Global_H3K27me3_Levels.png")
```

Save and exit (Ctrl+O, Enter, Ctrl+X)

Step 3: Run the Script

```
Rscript plot_minute.R
```


Global H3K27me3 Quantification
MINUTE-ChIP Scaled Ratios (Relative to Naive Control)
Scaled Fraction (MSR)
1.00
0.75
0.50
0.25
0.00
1
0.03
0.31
0.07
Naive (Control)
Naive + EZH2i
Primed
Primed + EZH2i